

Jean Pierre Twagirayezu, PhD

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Education

Michigan State University *Jan 2020 – May 2026*

Ph.D. in Physics — Neutrino Astronomy

Michigan State University *Aug 2018 – Dec 2020*

M.S. in Physics

African Institute for Mathematical Sciences *Sep 2017– June 2018*

M.S. in Mathematical Sciences

University of Rwanda, College of Education *Sep 2011 – July 2015*

Bachelor of Education (Hons.) in Physics

Research Work Experience

Graduate Research Assistant *East Lansing, MI, USA*

Michigan State University, Pacific Ocean Neutrino Experiment *Aug 2021 – Dec 2025*

- collaborated with a team of software engineers and scientists to study the performance of the P-ONE using Monte Carlo simulations.
- conducting statistical data analysis to forecast the sensitivity of the P-ONE to astrophysical point sources of neutrinos
- co-development of track event reconstruction algorithms for the P-ONE.
- processing large datasets of Monte Carlo simulations from neutrino event generation to detector response simulation for the P-ONE using high-performance computing clusters
- skills: statistical data analysis, software development (Python, git) · scientific computing, teamwork, and presentation. Member of the P-ONE collaboration.

Graduate Research Assistant *East Lansing, MI, USA*

Michigan State University, IceCube Neutrino Observatory *May 2020 – Aug 2021*

- uses the likelihood method and photon propagation for energy reconstruction of cascade events from the IceCube Monte Carlo Simulation dataset on high-performance computing clusters and GPUs.
- member of the IceCube Collaboration, IceCube monitoring shift for data quality and detector operation.

Teaching Work Experience

Graduate Teaching Assistant *East Lansing, MI, USA*

Michigan State University, Department of Physics and Astronomy *Jan 2026 – May 2026*

- served as a teaching assistant for PHY 222 Studio Physics for Life Scientists II
- assisted with in-class group problem solving through integrated physics laboratory and discussion, grading lab reports, office hours through help room sessions, and recitation.

Graduate Teaching Assistant *East Lansing, MI, USA*

Michigan State University, Center for Integrative Studies in General Science *Jan 2019 - May 2020*

- Teaching and grading three sections of Introductory Physics (Lab), ISP 209L.

Graduate Teaching Assistant *University Park, PA*

Michigan State University, Physics and Astronomy *Aug 2018 – Dec 2018*

- Conduct help room sessions, assist in class demo setup for PHY 232 Introductory Physics II

Physics Teacher*Kagarama Secondary School**Kigali, Rwanda**Feb 2017 - Aug 2017*

- Taught upper and lower secondary physics courses. Designed and delivered lessons, conducted laboratory sessions, and designed assessments.

Physics Teacher*Saint Paul International School (SPIS)**Kigali, Rwanda**Jan 2016 - Aug 2017*

- Taught Cambridge As & A Level Physics courses. Designed and delivered course lessons, conducted physics laboratory sessions, and designed assessments.

Physics Teacher*APRED Ndera Secondary School**Kigali, Rwanda**July 2015 - Jan 2016*

- Taught upper secondary Physics courses. Designed and delivered course lessons, conducted physics laboratory sessions, and designed assessments.

Physics Teacher*Nu-Vision High School (N.V.H.S)**Kigali, Rwanda**Sep 2014 – Jul 2015*

- Taught Lower secondary school Physics classes. Designed and delivered course lessons and designed assessments.

Mentoring Experience

Mentoring Undergraduate Students*Michigan State University, the P-ONE Group**East Lansing, MI, US**May 2023 - May 2025*

- I have mentored two Undergraduate students working in the MSU P-ONE Research Group to work on different projects, ranging from using computing clusters to produce Monte Carlo simulations, development of neutrino event track reconstruction using Monte Carlo simulations, and performing P-ONE geometry optimization studies.

Mentoring High School*UN Ivy STEM Connect - AIMS Rwanda, Lycée de Kigali**Kigali, Rwanda**Dec 2017 – Apr 2018*

- Provided mentorship to students from Lycée de Kigali, a secondary school, in Mathematics and Physics with AIMS Rwanda students and Ivy League students studying in the USA.

Technical Skills

Languages: English (Fluent), French (Intermediate), Kinyarwanda (Native)

Programming tools: Python, SQL, C++ (basic), shell (basic), Icecube icetray, Latex, Excel.

Data Analysis tools: Numpy, Pandas, Scipy, Sklearn, Keras, Matplotlib, Jupyter Notebook, Git/Github

Research & Teaching: Statistical Data Analysis, Machine Learning, Advanced Maths, Astroparticle Physics, Software development, Teaching, Communication, Presentation, Team work.